

Statement of Work. Statement of work for Preventive Maintenance and On-Site Repair of Uninterruptible Power Supplies, Power Distribution Units, and Direct Expansion InRow Cooling manufactured by APC/Schneider Electric.

Location: Naval Base Ventura County, [REDACTED], Naval Air Warfare Center – Weapons Division, Point Mugu CA, 93042-5035 USA.

1.0 Scope

- 1.1 This statement of work defines the effort required for preventive maintenance and on-site remedial visits for one uninterruptible power supplies manufactured by APC/Schneider Electric – an APC Symmetra PX 50kW 208V with startup serial number: QD1917240319; InRow RD 500mm Air Cooled 200-240V 50-60Hz with Humidifier serial number JK1831002838 and JK1831002841; Symmetra-LX SYA8K16PXR serial number: AD1837320469, any software updates and fixes according to manufacturer’s specifications.

2.0 Requirements. Symmetra PX 50kW 208V with Startup, S/N: QD1917240319.

- 2.1 General. The contractor shall provide a specialized labor, OEM Materials and OEM Parts to repair, diagnose, and maintain the APC Symmetra PX 50kW in [REDACTED] at the Naval Air Warfare Center – Weapons Division, Point Mugu CA. The service shall include the repair, diagnosis, and maintenance of the hardware indicated in Section 1.0. The service shall only be conducted by qualified Schneider Electric Critical Power and Cooling Services certified personnel. The service shall include repair parts costs, internal batteries, labor, and travel expenses. The service shall include access to mySchneider Portal and 24 hour – 7 days a week remote technical support. The service shall include firmware updates, as needed. The service shall include access to “EcoStruxure IT Free”.

- 2.1.1 *Annual Preventive Maintenance.* The contractor shall provide one (1) scheduled preventive maintenance service visit performed by Schneider Electric Critical Power and Cooling Services certified personnel during the yearly agreement for the hardware indicated in Section 2.0.

- 2.1.1.1 The contractor shall provide a Schneider Electric Critical Power and Cooling Services certified personnel to perform the work on the Uninterruptible Power Supply and/or Power Distribution Unit indicted in Section 1.0. and provide a printed document summarizing the results.
 - 2.1.1.2 The contractor shall perform visual inspections. Inspect the UPS and/or PDU solution to verify that system components are clean, (non-interventional only and will be specific to exterior and dust and debris only), and functioning within designed specifications.

- 2.1.1.3 The contractor shall perform Environmental Inspection. Verify and document that the system's environment is within specified operating conditions including but not limited to room temperature, airflow, dust contamination, etc.
- 2.1.1.4 The contractor shall perform Mechanical / Electrical Inspection. Inspect power and control wire termination points as well as Uninterruptible Power Supply PS and/or Power Distribution Unit. Perform Functional Verification Check and/or Power Distribution Unit event and alarm logs.
- 2.1.1.5 The contractor shall verify that input, output and bypass voltage and current values are within designed specifications, when applicable for the system configuration. Perform battery self-test, when applicable for the system configuration. Verify transfer to battery operation and transfer to and from static bypass, when applicable for the system configuration. Check parallel operation performance, when applicable for the system configuration.
- 2.1.1.6 The contractor shall implement updates. Verify and implement required Field Advisories and Field Modifications. Check circuit board revisions and upgrade as required. Update the firmware of the equipment as recommended by the manufacturer.
- 2.1.1.7 The contractor shall deliver documentation. Deliver a graphical site report documenting UPS and/or PDU status and on-site activities. Recommend any additional service activities as required resulting from the Preventive Maintenance activities listed above

2.1.2 *On-Site Remedial Visit.* As needed. During these service visits the contractor shall perform the following tasks.

- 2.1.2.1 The contractor shall provide Schneider Electric Critical Power and Cooling Services representatives at the customer's location to arrive next business day to diagnose, repair and test the system when required. Related labor, travel and parts shall be included with this service.
- 2.1.2.2 The contractor shall check UPS and/or PDU Status. Document the status of the Uninterruptible Power Supply (UPS) and/or Power Distribution Unit (PDU) upon arrival to the site, such as "on-line", "on-battery", "bypass" or other.
- 2.1.2.3 The contractor shall Check UPS Alarms. View event logs and displays for alarms / information on the UPS.

- 2.1.2.4 The contractor shall troubleshoot reported issue as required.
- 2.1.2.5 The contractor shall repair the system and replace parts with OEM materials and OEM parts as required in accordance with the manufacturer's specifications
- 2.1.2.6 The contractor shall perform functional tests conducted after corrective action is taken.
- 2.1.2.7 The contractor shall prepare and deliver a written report. Describe the problem and explain the corrective action taken. A detailed report will be provided.

3.0 Requirements. Two (2) InRow RD 500mm Air Cooled 200-240v 50-60Hz with Humidifier, Serial Numbers: JK1831002838 and JK1831002841

3.1 General. The contractor shall provide a specialized labor, OEM Materials and OEM Parts to repair, diagnose, and maintain two (2) inRow RD 500mm Air Cooled 200-240v 50-60hz with Humidifier units in [REDACTED] at the Naval Air Warfare Center – Weapons Division, Point Mugu CA. The service shall include the repair, diagnosis, and maintenance of the hardware indicated in Section 1.0. The service shall only be conducted by qualified Schneider Electric Critical Power and Cooling Services certified personnel. The service shall include repair parts costs, internal batteries, labor, and travel expenses. The service shall include access to mySchneider Portal and 24 hour – 7 days a week remote technical support. The service shall include firmware updates, as needed. The service shall include access to “EcoStruxure IT Free”.

3.1.1 *Preventive Maintenance.* The contractor shall provide TWO (2) visits per year per system performed by a Schneider Electric Critical Power and Cooling Services certified technician during the yearly agreement for the InRow RD hardware indicated in Section 3.0.

3.1.1.1 The contractor shall perform environmental inspection. Measure and record room temperature and humidity, check the system for adequate cooling capacity to support the load and make recommendations as necessary, ensure the environment is within manufacturer specified operating conditions and clearances and document any environmental noncompliance issues and recommend appropriate action as necessary.

3.1.1.2 The contractor shall perform the following maintenance tasks, as applicable.

3.1.1.2.1 Inspect unit for damage, missing or broken hardware, and abnormal noises. Inspect doors, panels, gaskets, and hardware.

3.1.1.2.2 Check external paint integrity. Inspect for foreign debris and signs of corrosion within the unit.

- 3.1.1.2.3 Check and tighten electrical connections within the unit. Replace drive belts, check and lubricate bearings, check motor mounts, pulleys, and set screws.
- 3.1.1.2.4 Clean or replace the factory installed air filter and water filter media. Inspect heat exchangers, fan blades, and coil fins. Inspect for water leaks on the factory supplied piping system.
- 3.1.1.2.5 Visual inspection for refrigerant and or Chilled Water leaks.
- 3.1.1.2.6 Visually inspect for Water/Glycol condenser loop for leaks if applicable.
- 3.1.1.2.7 Inspect condensate pan and verify proper condensate removal from unit.
- 3.1.1.2.8 Clean or replace the humidifier cylinder. Inspect humidifier water valves.
- 3.1.1.2.9 Check and tighten pipe clamps, piping joints, and compressor mounting hardware within the unit.
- 3.1.1.3 The contractor shall check system operating conditions by performing the following tasks.
- 3.1.1.4 Measure and record the primary and secondary voltages and major component amperage readings.
- 3.1.1.5 Check and adjust fan speed regulation and controls.
- 3.1.1.6 Check the chilled water and/or condenser water supply temperature and water flow. Check the glycol concentration.
- 3.1.1.7 Verify operation of the water regulating valve and actuator if applicable.
- 3.1.1.8 Check compressor operation, if applicable.
- 3.1.1.9 Measure and record refrigerant pressures and temperatures.
- 3.1.1.10 Check the refrigerant and compressor oil level; check the filter drier pressure drop.
- 3.1.1.11 Check the refrigerant metering device and head pressure control valve operation.
- 3.1.1.12 Check operation of the Outdoor Condenser, Fluid Cooler, and Pump Package if applicable.
- 3.1.1.13 Check and adjust fan speed and pump control settings.
- 3.1.1.14 The Contractor shall verify unit modes of operation: (Cooling/Reheat/Humidification/Dehumidification) by completing the following tasks:
 - 3.1.1.14.1 Confirm unit's ability to maintain temperature and humidity set-points.
 - 3.1.1.14.2 Verify controller configuration and control setpoints.

- 3.1.1.14.3 Check and calibrate safety switches, pressure, temperature, and humidity sensors.
- 3.1.1.14.4 Check operation of group control, redundancy failover, and lead-lag functions.
- 3.1.1.14.5 Review Alarm history and investigate logged alarms.
- 3.1.1.14.6 The contractor shall provide documentation.
Document system condition and further service needs and provide that document to the customer.
- 3.1.1.14.7 Make recommendations to the customer regarding cooling solution repairs or enhancements, if required.

3.1.2 *Onsite Remedial Service.* As needed. During these service visits the contractor shall perform the following tasks.

- 3.1.2.1 The contractor shall provide Schneider Electric Critical Power and Cooling Services representatives at the customer's location to arrive next business day to diagnose, repair and test the system when required. Related labor, travel and parts shall be included with this service.
- 3.1.2.2 Cooling equipment status check. Document the status of the cooling unit upon arrival to the site (i.e., verify alarms, loss of cooling, operating parameters).
- 3.1.2.3 System alarms check. View active alarms and event logs. Download event logs from the cooling unit (if applicable).
- 3.1.2.4 Review alarm history and investigate logged alarms.
- 3.1.2.5 Diagnose. Troubleshoot reported issue if required. Perform environmental and operational inspection. Check system parameters and operating conditions.
- 3.1.2.6 Repair. Repair the system if required. Perform necessary adjustments to restore proper operation. Verify proper operation of control systems and safety devices.
- 3.1.2.7 Functional Test. Perform functional testing after corrective action is taken. Verify proper operation. Confirm system maintains required setpoints. Test group control and redundancy features (if applicable).
- 3.1.2.8 Report Preparation and Delivery. Describe the diagnosis and corrective action taken. Document any recommendations for system improvements. Provide a detailed service report including relevant measurements and observations.

3.1.3 *Customer Responsibilities*

- 3.1.3.1 Coolant Management (if applicable). Maintain adequate supply of manufacturer-approved coolant for routine top-offs and emergency situations;
- 3.1.3.2 Perform regular coolant testing and analysis as specified in the Operation and Maintenance Manual;
- 3.1.3.3 Provide coolant test reports to Schneider Electric Services Representatives prior to each maintenance visit;
- 3.1.3.4 Maintain proper chemical treatment and inhibitor levels as specified by coolant manufacturer;
- 3.1.3.5 Properly dispose of used or contaminated coolant according to local regulations;
- 3.1.3.6 Maintain documentation of coolant maintenance history including:
 - 3.1.3.6.1 Test results, top-off dates and quantities, chemical treatment dates, and replacement dates
 - 3.1.3.6.2 Immediately notify OEM of any observed coolant-related issues (leaks, discoloration, etc.).

4.0 Requirements. Symmetra LX - SYA8K16PXR, SN: AD1837320469.

4.1 General. The contractor shall provide a specialized labor, OEM Materials and OEM Parts to repair, diagnose, and maintain and conduct a pre-contract audit for the Symmetra LX in [REDACTED] at the Naval Air Warfare Center – Weapons Division, Point Mugu CA. The service shall include the repair, diagnosis, and maintenance of the Symmetra LX hardware indicated in Section 1.0. The service shall only be conducted by qualified Schneider Electric Critical Power and Cooling Services certified personnel. The service shall include repair parts costs, internal batteries, labor, and travel expenses. The service shall include access to mySchneider Portal and 24 hour – 7 days a week remote technical support. The service shall include firmware updates, as needed. The service shall include access to “EcoStruxure IT Free”.

4.1.1 *Onsite Remedial Service.* As needed. This service is intended for single phase UPS and/or PDUs performed by qualified Schneider Electric Services personnel at the Customer’s location to arrive next business day to diagnose, repair, and test the system in the event that this does not function as expected. The contractor shall perform the service tasks detailed below during this visit:

- 4.1.1.1 Check UPS Status
- 4.1.1.2 Document the status of the UPS and/or PDU upon arrival to the site (i.e. on-line, on-battery, static bypass, or other, etc.)
- 4.1.1.3 Check UPS Alarms
- 4.1.1.4 View event logs and display for alarms / information on both UPS/PDUs and in static bypass mode.
- 4.1.1.5 Diagnose. Troubleshoot reported issue as required.

- 4.1.1.6 Repair . Replace any parts that is not functioning as expected and repair the system using OEM parts and materials in accordance with manufacturer's specifications as required.
- 4.1.1.7 Test. Complete functional tests conducted after corrective action is taken.
- 4.1.1.8 Prepare and Deliver Report. Describe the problem and describe the corrective action taken. A detailed report will be provided.

4.1.2 *Pre-contract Audit.* The contractor shall provide a visual, environmental and electronic inspection and baseline report of the UPS and/PDU to define the current status of critical systems. The contractor shall provide Schneider Electric Critical Power and Cooling Services certified personnel to perform the following tasks:

- 4.1.2.1 Visual Inspection. Inspect and document the interior and exterior condition of the UPS and/or PDU solution to generate the audit report.
- 4.1.2.2 Environmental Inspection. Inspect and document the system's environmental operating conditions including but not limited to datacenter design, location of equipment, room temperature, airflow, dust contamination, etc.
- 4.1.2.3 Mechanical/Electrical Inspection. Inspect and document the condition of power and control wiring termination points as well as all UPS and/or PDU system components.
- 4.1.2.4 Functional Verification. Check the different modes of operation and record the result of the functional verification of the UPS and/or PDU event as well as the alarm logs. The modes of operation include verification of input, output and bypass voltage and current values are within designed specifications, when applicable to the system configuration.
- 4.1.2.5 Verify transfer to on battery operation and transfer to and from static bypass, when applicable to the system configuration.
- 4.1.2.6 Check parallel operation performance, when applicable to the system configuration.
- 4.1.2.7 Status Verification of Installed Upgrades, Enhancements and Service Bulletins
- 4.1.2.8 Document the different product enhancements, the safety upgrades, the service bulletins and firmware revision currently installed on the system.
- 4.1.2.9 System Audit Report. Following the on-site visit, Provider will forward onto the Customer a written audit detailing the

current status of their system versus the minimum requirements for contract eligibility.

- 4.1.2.10 Remedial Repair Estimate. Following the on-site visit, provider will forward onto the Customer a time and materials estimate for the repair work required to meet the minimum requirements for contract eligibility.